

Porcine Sclera: Thickness and Surface Area

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PURPOSE. To assess the thickness and surface area of porcine sclera.

METHODS. One hundred twenty-eight porcine globes were sectioned from the center of the cornea to the region of the optic nerve. Photographs of the sectioned sclera including a millimeter scale were taken. Photographic slides were projected onto blank paper, and the scleral silhouette was traced. Perpendicular thickness measurements were taken at 1-mm intervals from the limbus to the optic nerve. The sclera of 18 porcine eyes were cut into small pieces, and the surface area was calculated with computerized digital tracing software.

RESULTS. The scleral thickness near the corneal scleral limbus was 0.83 ± 0.2 , 0.91 ± 0.17 , and 1.12 ± 0.23 mm in the small-, medium-, and large-sized pigs, respectively. Thickness decreased to minimum of 0.31 ± 0.07 , 0.35 ± 0.1 , and 0.43 ± 0.16 mm at a distance of 5 mm from the limbus in the small- and medium-sized pigs and 6 mm in the large-sized pigs, respectively. The mean scleral surface area was 7.78 ± 0.66 , 9.66 ± 0.75 , and 11.92 ± 1.57 cm² in the small-, medium-, and large-sized pigs, whereas the corneal surface area was 1.09 ± 0.07 , 1.15 ± 0.09 , and 1.40 ± 0.19 cm², respectively.

CONCLUSIONS. Porcine scleral thickness is very similar to human scleral thickness. The porcine model is an excellent model for studying transscleral drug delivery. (*Invest Ophthalmol Vis Sci*. 2002;43:2529–2532)

Isolated human sclera has been shown to be permeable to various hydrophilic compounds in vitro.¹ Small hydrophilic molecules diffuse readily through the sclera. Large dextran polymers with molecular weights as high as 70,000 kDa also diffuse through sclera. Ambati et al.² have demonstrated that large molecules, such as IgG with molecular weights as large as 150 kDa could diffuse across fresh rabbit sclera in vitro and suggested that molecular radius is a good predictor of scleral permeability. Other bioactive molecules, such as the monoclonal antibody and anti-intercellular adhesion molecule (ICAM)-1, were delivered in vivo in a rabbit model of transscleral drug delivery.³ We have documented that the mean sclera thickness in a human eye is approximately 0.5 mm at the corneal scleral limbus, decreasing to approximately 0.4 mm near the equator and increasing to 1 mm near the optic nerve head.⁴ We believe that the porcine eye is an excellent animal model for studying transscleral drug delivery based primarily on the scleral thickness. Other advantages of the porcine model include holangiotic retinal

vasculature, choroidal blood flow, a retinal pigment epithelium that is similar to human, the absence of a tapetum, and the presence of cone photoreceptors in the outer retina.⁵

METHODS

Porcine globes ($N = 128$) were obtained from 65 domestic pigs (*Sus scrofa domestica*) that ranged in weight between 2.8 and 81 kg. We separated the pigs into three groups based on weight: small (≤ 8 kg, $n = 38$), medium (8.1–26.9 kg, $n = 40$), and large (≥ 27 kg, $n = 42$). Animals were killed for various non-ocular, research-related reasons, but none was killed specifically for the purpose of this study. Most of the eyes were collected from the veterinary diagnostics laboratory at the University of Minnesota. All eyes were enucleated within 2 hours of death and fixed with 10% phosphate-buffered neutral formalin. Eyes were sectioned from 1 to 4 weeks from the time of collection, by using a straight razor blade from the center of the cornea in an anterior-posterior direction, transecting the lens and posterior pole region near the optic nerve. Every attempt was made to transect each globe at its greatest diameter. High-resolution color photographs were taken with a photographic stand, and the perpendicular cut edge of the sclera was photographed with a millimeter scale that was held at precisely the level of the cut sclera (Fig. 1). Photographic slides of the sectioned sclera were projected onto a blank white sheet of paper. The distance of projection was adjusted so that 1 mm on the ruler scale corresponded with 1 cm on the projected image. The scleral silhouette was carefully traced on the white paper with a sharp pencil. Perpendicular thickness measurements were taken at 1-cm increments from the limbus to the posterior pole with the intervals measured with a ruler (accurate to ± 0.5 mm). The limbal area of one globe was sectioned and sent for histology with hematoxylin-eosin staining (Fig. 2).

Porcine eyes ($n = 18$; six from each group) were sectioned as described. The uveal tissue was removed, and the sclera was cut into small pieces (32–40 per globe) and photographed (Fig. 3). The surface of each cut piece of scleral tissue was flattened, without folding or distortion, by placing them on an acrylic sheet and covering them with a glass slide to keep the surface area in a single plane. Tissue was photographed using a flat copy stand, and high-resolution color slides were scanned with a film scanner (LS-2000; Nikon, Melville, NY). Images were then imported into image software (NIH ver.1.62; NIH Image, W. Rasband, National Institutes of Health; available by ftp from zippy.nimh.nih.gov or on floppy disc from NTIS, Springfield, VA, part number PB95-500195GED). Outlines were traced and the surface area calculated.

RESULTS

Scleral thickness near the corneal scleral limbus was 0.83 ± 0.20 , 0.91 ± 0.17 , and 1.12 ± 0.23 mm in the small-, medium-, and large-sized pigs. The thinnest area occurred at a distance of 5 mm from the limbus in the small- and medium-sized pigs with measurements of 0.31 ± 0.07 and 0.35 ± 0.1 mm, respectively. The thinnest area of the large pig was 6 mm posterior to the limbus with a measurement of 0.43 ± 0.16 mm (Table 1). In the small animals, the mean scleral thickness near the equator was 0.56 ± 0.11 mm, whereas in the medium-sized pigs, the thickness increased to 0.73 ± 0.14 mm and in the large pigs, to 0.86 ± 0.18 mm. In the more posterior measurements, the scleral thickness increased until approximately 10 to 12 mm posterior to the limbus, and then remained at a constant thick-

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Supported by the Minnesota Lions Macular Degeneration Center, Minneapolis, Minnesota; the University of Minnesota Vision Foundation; and Research to Prevent Blindness.

Submitted for publication January 25, 2002; revised March 15, 2002; accepted March 29, 2002.

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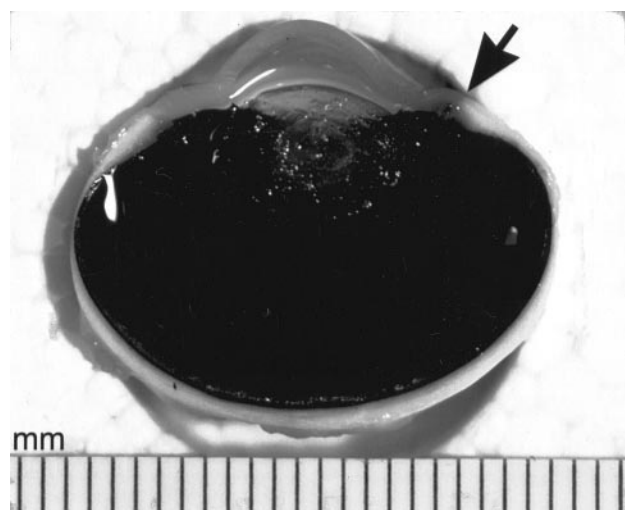


FIGURE 1. Cross-section of porcine globe after fixation. A millimeter scale is situated inferiorly. Arrow: the limbus and starting point for measurement of scleral thickness.

ness toward the optic nerve (Fig. 4). Mean scleral surface area in the small-, medium-, and large-sized pigs was 7.78 ± 0.66 , 9.66 ± 0.75 , and 11.92 ± 1.57 cm², respectively. Mean corneal

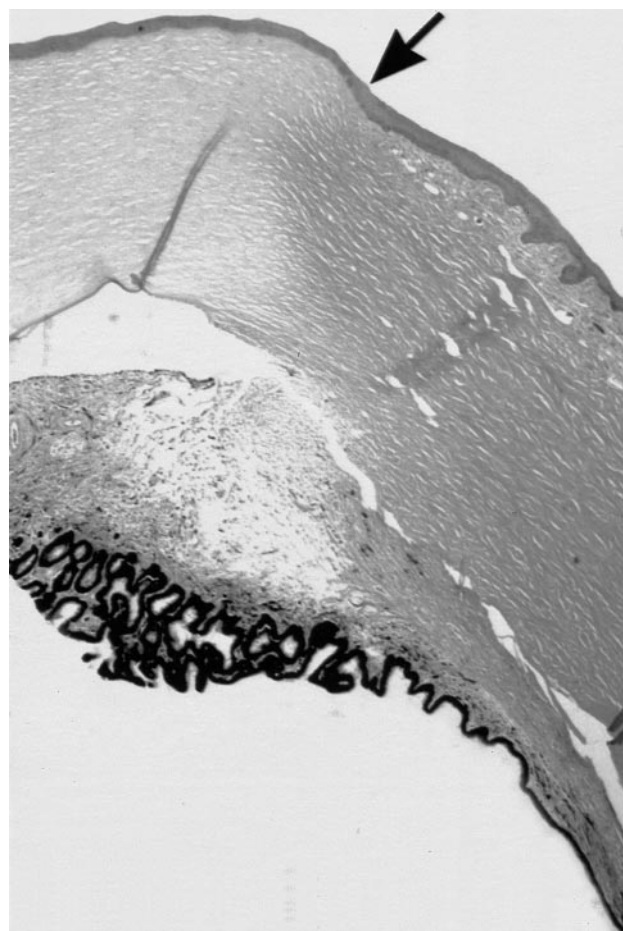


FIGURE 2. Histology of the limbus. Arrow: the limbus. Corneal stroma appears to the left, and scleral tissue to the right of the arrow. The pigmented ciliary body attaches near the inner surface of the limbus. Hematoxylin-eosin; original magnification, $\times 10$.

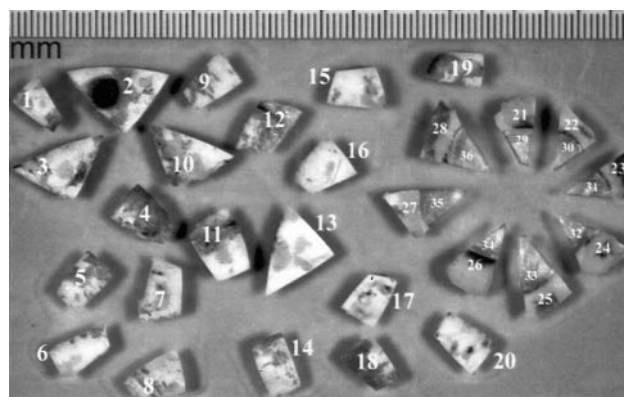


FIGURE 3. Multiple scleral segments from a single porcine globe with a millimeter scale alongside. Specimens were flattened under acrylic and subsequently traced with surface area software.

surface area was 1.09 ± 0.07 , 1.15 ± 0.09 , and 1.40 ± 0.19 cm², respectively (Fig. 5). Mean human corneal surface area is 1.3 cm².⁶

DISCUSSION

The pig is an excellent model for studying the pharmacokinetics of transscleral drug delivery in vivo. There are many advantages over other animal models. The porcine sclera has thickness measurements that are very similar to those of humans. The thinnest area of the human sclera is approximately 0.4 mm, near the equator (13 mm posterior to the limbus).⁴ This corresponds very closely to the thin region in the porcine eye, located approximately 5 to 6 mm posterior to the limbus. In vitro studies have demonstrated that scleral permeability is inversely related to scleral thickness.¹ To increase drug diffusion, device placement 5 to 6 mm from the limbus would target thinner regions of the sclera. Placement of a drug delivery system on the anterior region of the porcine eye, or 5 to 6 mm from the limbus, would be analogous to placement of the device near the equator of a human eye. There are important structural differences between the ocular adnexa of the pig and that of human globes. Anterior localization of extraocular devices in a pig avoids the dense bands of extraocular musculature that surround the globe. In primates, the extraocular musculature is less of an impediment to placement of extraocular material. For example, scleral buckles are commonly placed around the globe with infrequent long-term consequences.

Other animal models do not have similar scleral thickness measurements. For example, the scleral thickness of rabbit eyes is 0.2 to 0.25 mm near the equator.⁷ Similarly, the scleral thickness at the foveola of subhuman primates (*Macaca fascicularis*) is approximately 0.26 mm.⁸ The total scleral surface area is smaller in pigs than in humans. The mean scleral surface area of a human globe⁴ is approximately 16 to 17 cm², whereas the sclera of a small pig is approximately 8 cm² and increases to 12 cm² in the larger animal.

The intraocular anatomy of the porcine eye is more analogous to that of the human eye in several respects. First, the retinal vascular pattern is holangiotic (fully vascularized) as opposed to the rabbit, in which the retina is merangiotic, or mostly avascular. The retinal pigment epithelium is more analogous anatomically and structurally. Many animals, such as cats and dogs, have a tapetum that makes imaging of the posterior pole difficult and could alter diffusion kinetics. The pig does not have a tapetum. The choroidal vasculature and Bruch's membrane of the pig are analogous to that of the human.⁵

TABLE 1. Scleral Thickness and Surface Area in the Pig

	Eyes (n)	Scleral Thickness (mm)		Scleral Surface Area (cm ²)
		Thinnest	At the Equator	
Small	38	0.31 ± 0.07*	0.56 ± 0.01	7.78 ± 0.66
Medium	40	0.35 ± 0.10*	0.73 ± 0.02	9.66 ± 0.75
Large	42	0.43 ± 0.13†	0.86 ± 0.02	11.92 ± 1.57

Data are expressed as the mean ± SD.
* Thinnest area is 5 mm from the limbus.
† Thinnest area is 6 mm from the limbus.

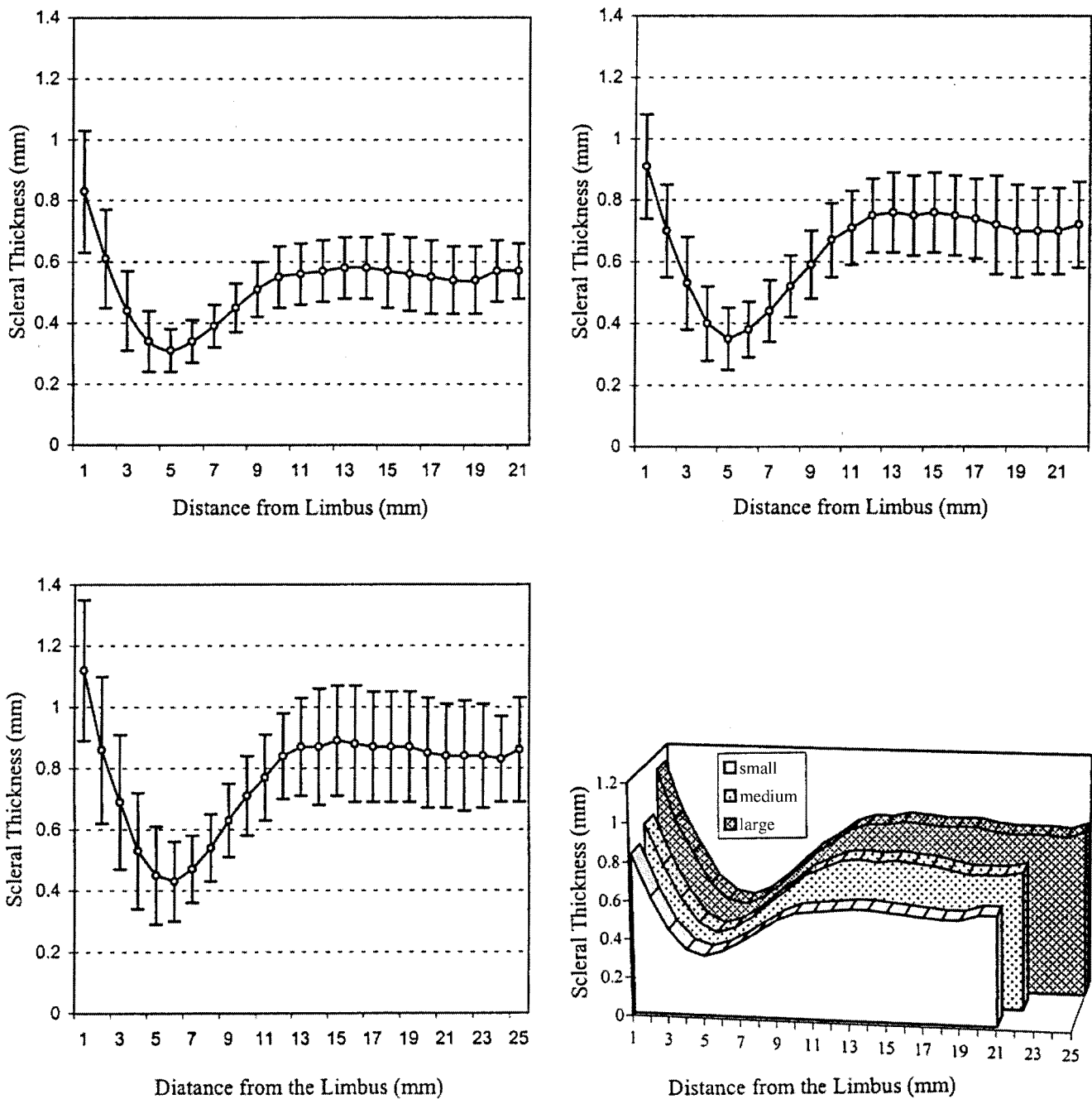


FIGURE 4. Comparison of scleral thickness with distance from the limbus in small (top left), medium (top right), and large (bottom left) porcine eyes (±SD). A composite three-dimensional comparison is shown (bottom right).

Finally, variability of the scleral ultrastructure may also introduce variability in the diffusion characteristics of the sclera.

Disadvantages of the porcine model include the relatively rapid growth of the animals that makes long-term studies difficult. The orbit is small, and surgical placement of extraocular devices is challenging. Porcine extraocular musculature is extremely robust, and devices are difficult to place posterior to the muscle insertions, particularly inferiorly. Although pigs do not have a true macula, they have an area centralis with cone photoreceptors in a region that mirrors the primate macula anatomically.⁵ We have also noted an anterior, circumferential retinal blood vessel that travels parallel to the ora serrata (Fig. 6). This vessel could introduce an additional variable in the study of drug kinetics.

In summary, the anatomic structure of the porcine sclera as well as the intraocular choroidal blood supply, retinal pigment epithelium, retinal vasculature, area centralis with cone photoreceptors, and absence of a tapetum are advantageous structural features that make the porcine model ideal for studying transscleral drug diffusion in vivo. Other animal models have limitations based on scleral thickness, retinal blood flow patterns, choroid, retinal pigment epithelium, the presence of a tapetum, and the type of photoreceptors present. Defining the anatomy, thickness, and surface area of the porcine sclera is an important step in developing an in vivo model for transscleral drug diffusion. A better understanding of the distinct differ-

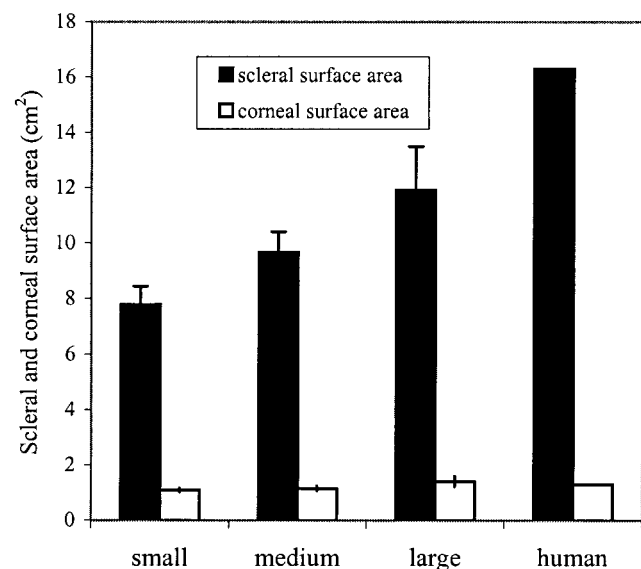


FIGURE 5. Comparison of mean small, medium, and large porcine scleral and corneal surface areas to that of humans ($\pm$ SD).^{4,5}

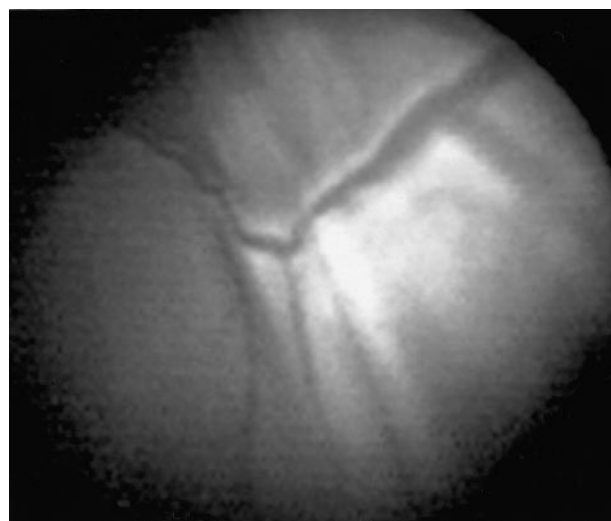


FIGURE 6. Circumferential vessel at the ora serrata in a porcine eye. The tissue superior to the vessel is the pars plana, and the tissue inferior to the vessel is the vascularized retina (note radial vessels).

ences between human and porcine ocular anatomy will be helpful in testing transscleral drug delivery systems.

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